Safety Data Sheet according to UN-GHS

1. Identification

Product identifier: **MERACRYL® HEMA 98**

Chemical name: 2-Hydroxyethyl methacrylate

Other means of identification

Recommended use: Monomer Chemical Intermediate Consumers professional user

Recommended restrictions: Applications where liquid monomer is intended to come into contact with skin or nails.

Manufacturer/Importer/Distributor Information

Company Name	:	Roehm Hong Kong Co., Ltd. Unit 1504-05A, 15/F, 909 Cheung Sha Wan Road Cheung Sha Wan, Kowloon Hong Kong
Telephone	:	+86 21 6759 1000
E-mail	:	SDS-PS-Asia@roehm.com

Emergency telephone number:

24-Hour Health Emergency	:	+86 532 8388 9090 (仅中国大陆 / China mainland only) +49 6241 402 5280 (24h)

2. Hazard(s) identification

Classification according to GHS

Health Hazards

Serious Eye Damage/Eye Irritation	Category 2B
Skin sensitizer	Category 1

Label Elements

Hazard Symbol:

Signal Word: Warning

Hazard Statement: Causes eye irritation.
May cause an allergic skin reaction.

Precautionary Statements

Prevention: Avoid breathing dust/fume/gas/mist/vapors/spray. Wash face, hands and any exposed skin thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace. Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response: IF ON SKIN: Wash with plenty of soap and water. If skin irritation or rash occurs: Get medical advice/attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention. Take off contaminated clothing and wash it before reuse.

Disposal: Dispose of contents/container to an appropriate treatment and disposal facility in accordance with applicable laws and regulations, and product characteristics at time of disposal.

Other hazards: Take precautionary measures against static discharges. Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions.

3. Composition/information on ingredients

Substances

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%)*
2-hydroxyethyl methacrylate	2-Propenoic acid, 2-methyl-, 2-hydroxyethyl ester	868-77-9	>=98%

* All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

Composition information of impurities and stabilizers

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%)*
diethyleneglycol monomethacrylate	2-(2-Hydroxyethoxy)ethyl methacrylate	2351-43-1	0,5 - 1,5%
Ethylene glycol dimethacrylate	2-[(2-methylprop-2-enoyl)oxy]ethyl 2-methylprop-2-enoate	97-90-5	0,1 - <1%

* All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

The exact concentration has been withheld as a trade secret.

4. First-aid measures

Description of first aid measures

General information: First aider needs to protect himself. Take off all contaminated clothing immediately. Medical treatment is necessary if symptoms occur which are obviously caused by skin or eye contact with the product or by inhalation of its vapours.

Inhalation: Move subject to fresh air and keep him calm. Get medical attention if any discomfort continues.

Skin Contact: Wash off immediately with soap and water. If skin irritation occurs consult a physician.

Eye contact: Immediately flush with plenty of water for up to 15 minutes. Remove any contact lenses and open eyes wide apart. Consult a physician.

Ingestion: Do not induce vomiting. Call a physician immediately. Never give anything by mouth to an unconscious person.

Personal Protection for First-aid Responders: First Aid responders should pay attention to self-protection and use the recommended protective clothing

Most important symptoms and effects, both acute and delayed

Symptoms: Causes eye irritation. Skin sensitizer

Hazards: Causes eye irritation. May cause an allergic skin reaction.

Indication of immediate medical attention and special treatment needed

Treatment: Treat symptomatically.

5. Fire-fighting measures

General Fire Hazards: Standard procedure for chemical fires. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Combustible liquid. Evacuate personnel to safe areas. Keep out unprotected persons. Use water spray to cool unopened containers. Closed container may rupture if strongly heated. Move containers from fire area if you can do so without risk. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. Prevent fire extinguishing water from contaminating surface water or the ground water system.

Suitable (and unsuitable) extinguishing media

Suitable extinguishing media: In case of fire: Use water spray, foam, dry chemical powder or carbon dioxide to extinguish.

Unsuitable extinguishing media: High volume water jet

Special hazards arising from the substance or mixture: Closed container may rupture if strongly heated. May be released in case of fire: carbon monoxide, carbon dioxide, organic products of decomposition.

Special protective equipment and precautions for fire-fighters

Special fire-fighting procedures: Take action to prevent static discharges. In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitible mixtures may form in air. Vapours are heavier than air and may spread along floors.

Special protective equipment for fire-fighters: Wear self-contained breathing apparatus.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures: Assure sufficient ventilation. Use personal protective clothing. Keep away sources of ignition. Use breathing apparatus if exposed to vapours/dust/mist/aerosol. Do not breathe vapours or spray mist. Do not eat, drink or smoke when using this product. Wash hands thoroughly with soap and water after handling.

Accidental release measures: Evacuate area and do not approach spilled product. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). For personal protection see section 8.

For emergency responders: Avoid contact with eyes, skin, and clothing. Do not inhale vapours / aerosols. Observe regulations on prevention of water pollution (check, dam up, cover up).

Methods and material for containment and cleaning up: Larger quantities: Take up mechanically (by pumping). Smaller quantities and/or residues: Contain with absorbent material (e.g. sand, diatomaceous earth, acid absorbent, universal absorbent or sawdust). Dispose of in accordance with regulations.

Environmental Precautions: Prevent product from getting into drains/surface water/groundwater. If the product contaminates rivers and lakes or drains inform respective authorities.

7. Handling and storage

Handling

Technical measures: Install appropriate equipment and wear appropriate personal protective equipment (see "8. Exposure control/personal protection").

Local/Total ventilation: Provide adequate ventilation.

Safe handling advice: Handle in accordance with good industrial hygiene and safety practice. Keep container tightly closed. Ensure there is good room ventilation. Use personal protective equipment.

Do not get in eyes, on skin, or on clothing.Do not breathe vapors. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Wash thoroughly after handling. Use only trained personnel.Keep away from sources of ignition - No smoking. Take action to prevent static discharges. In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitible mixtures may form in air. Vapours are heavier than air and may spread along floors. Follow all SDS/label precautions even after container is emptied because it may retain product residues.

Contact avoidance measures: see section 8. see section 10.

Storage

Safe storage conditions: Keep only in the original container at a temperature not exceeding 30 °C. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Protect from the action of light. Fill the container by approximately 90 % only as oxygen (air) is required for stabilisation. With large storage containers make sure the oxygen (air) supply is sufficient to ensure stability. Can polymerize with intense heat release.Observe prohibition against storing together! see also section 10.For further information, refer to our web site: www.meracryl.com.

Safe packaging materials: No data available.

8. Exposure controls/personal protection

Control Parameters

Occupational Exposure Limits

None of the components have assigned exposure limits.
Observe national threshold limit values.

Biological Limit Values

No biological exposure limits noted for the ingredient(s).

Appropriate Engineering Controls For monitoring procedures refer for instance to "Empfohlene Analysenverfahren für Arbeitsplatzmessungen", Schriftenreihe der Bundesanstalt für Arbeitsschutz and "NIOSH Manual of Analytical Methods", National Institute for Occupational Safety and Health Eye wash facilities and emergency shower must be available when handling this product.

Individual protection measures, such as personal protective equipment (PPE)

General information: No data available.

Eye/face protection: tightly fitting goggles

Hand Protection:	Material: butyl rubber gloves (minimal thickness 0.3 mm) Break-through time: 480 min Guideline: EN 374 Additional Information: Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time., The above mentioned hand protection is based on special knowledge of the chemical and the intended handling of this product, however, it still may not be suited for all workplaces. A qualified hazard assessment should be made prior to the onset of work in order to determine the suitability of gloves for specific working environments and processes., Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
Other:	Choose body protection according to the amount and concentration of the dangerous substance at the work place. On handling of larger quantities: face mask, chemical-resistant boots and apron
Respiratory Protection:	Breathing apparatus in case of high concentrations
Hygiene measures:	Follow the usual good standards of occupational hygiene. Store work clothing separately. Take off all contaminated clothing immediately. Clean skin thoroughly after work; apply skin cream. Wash contaminated clothing before reuse. Contaminated work clothing should not be allowed out of the workplace.

9. Physical and chemical properties

Information on basic physical and chemical properties

Appearance

Physical state:	liquid
Form:	liquid
Color:	Colorless
Odor:	ester-like
Odor Threshold:	No data available.
Freezing point:	-99 °C/-146 °F Method: OECD 102
Boiling Point:	213 °C/415 °F (1.013 hPa) Method: OECD 103
Flammability:	Combustible liquid. Not classified as flammable but will burn.

Upper/lower limit on flammability or explosive limits

Explosive limit - upper:	No data available.
Explosive limit - lower:	No data available.
Flash Point:	106 °C/223 °F

	Method: ASTM D 93 (Pensky-Martens Closed Cup)
Auto-ignition temperature:	375 °C/707 °F 1.013 hPa Method: DIN 51794
Decomposition Temperature:	Not applicable
pH:	neutral, in Water
Viscosity	
Dynamic viscosity:	No data available.
Kinematic viscosity:	3,42 mm2/s (40 °C/104 °F) 6,36 mm2/s (20 °C/68 °F), Method: DIN 51562
Flow Time:	No data available.
Solubility(ies)	
Solubility in Water:	> 100 g/l (20 °C/68 °F) miscible
Solubility (other):	miscible with most organic solvents
Partition coefficient (n-octanol/water):	0,42 (25 °C/77 °F) Method: OECD 107
Vapor pressure:	0,08 hPa (20 °C/68 °F) Method: OECD 104, dynamic method
Relative density:	No data available.
Density:	1,07 g/cm3 (20 °C/68 °F)
Bulk density:	Not applicable
Relative vapor density:	> 1 (20 °C/68 °F)
Other information	
Explosive properties:	Not explosive Not to be expected in view of the structure Vapours may form explosive mixtures with air
Self-ignition:	not spontaneously flammable in air at ambient temperature (not pyrophoric)

10. Stability and reactivity

Reactivity:	Polymerisation can occur.
Chemical Stability:	No decomposition if used as directed.
Possibility of hazardous reactions:	Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions.
Conditions to avoid:	Ultraviolet light. High temperature. The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.
Incompatible Materials:	Peroxides, amines, sulfur compounds, heavy metal ions, alkalis, reducing agents and oxidizing agents.

Hazardous Decomposition Products: None when used as directed.

11. Toxicological information

General information: The substance is rapidly metabolized

Information on likely routes of exposure

Inhalation:	No specific health warnings noted.
Skin Contact:	May cause an allergic skin reaction.
Eye contact:	May irritate eyes.
Ingestion:	No specific health warnings noted.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation:	No specific symptoms noted.
Skin Contact:	Prolonged or repeated contact may cause skin sensitization in susceptible individuals.
Eye contact:	Eye may become red, tear, and become painful.
Ingestion:	No specific symptoms noted.

Acute toxicity (list all possible routes of exposure)

Oral

Product:	Not classified for acute toxicity based on available data.
Components:	
2-hydroxyethyl methacrylate	LD 50, Rat, > 5.000 mg/kg, FDA directive, Own study
diethyleneglycol monomethacrylate	Not toxic after single exposure, Not classified for acute toxicity based on available data.
Ethylene glycol dimethacrylate	LD 50, Rat, > 5.000 mg/kg, FDA directive

Dermal

Product:	Based on available data, the classification criteria are not met.
Components:	
2-hydroxyethyl methacrylate	LD 50, Rabbit, > 5.000 mg/kg, (analogy)
diethyleneglycol monomethacrylate	Not toxic after single exposure, Not classified for acute toxicity based on available data.
Ethylene glycol dimethacrylate	LD 50, > 2.000 mg/kg, OECD 402, Not toxic after single exposure, (limit test)

Inhalation

Product:	An Expert Judgment stated that no classification is necessary based on present knowledge.
Components:	
2-hydroxyethyl methacrylate	Not toxic after single exposure, Not classified based on available information., Vapour Not toxic after single exposure, Not classified based on available

	information., Dust and mist
diethyleneglycol monomethacrylate	Not toxic after single exposure, Vapour, Not classified for acute toxicity based on available data. Not toxic after single exposure, Dust and mist, Not classified for acute toxicity based on available data.
Ethylene glycol dimethacrylate	Not toxic after single exposure, Vapour, Based on available data, the classification criteria are not met. Not toxic after single exposure, Dust and mist, Not applicable

Repeated dose toxicity

Product:	NOAEL Rat, Oral, 7 Weeks, 100 mg/kg
Components:	
2-hydroxyethyl methacrylate	NOAEL Rat, Oral, 7 Weeks, 100 mg/kg
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	NOAEL Rat, Male, Oral, 49 d, 100 mg/kg NOAEL Rat, female, Oral, 49 d, 300 mg/kg

Skin Corrosion/Irritation

Product:	non-irritant
Components:	
2-hydroxyethyl methacrylate	Non-Irritating, Draize Test, Rabbit, 24 h, Own study
diethyleneglycol monomethacrylate	Irritating.
Ethylene glycol dimethacrylate	Non-Irritating, FDA 1959 Draize, occlusive, Rabbit, 24 h

Serious Eye Damage/Eye Irritation

Product:	Eye irritant Category 2B (UN-GHS)
Components:	
2-hydroxyethyl methacrylate	Slightly irritating, Draize Test, Rabbit, Own study Eye irritant Category 2B (UN-GHS)
diethyleneglycol monomethacrylate	Irritating.
Ethylene glycol dimethacrylate	Non-Irritating, Draize, reevaluated acc. to OECD 405, Rabbit

Respiratory or Skin Sensitization

Product:	Skin Sensitisation Category 1 (UN-GHS) May cause sensitization by skin contact. Buehler Test, Guinea Pig, Non sensitising, Cases of sensitisation also observed in humans. Not classified for respiratory sensitization
Components:	
2-hydroxyethyl methacrylate	Buehler Test, GPMT, Guinea Pig, Non sensitising, Skin Sensitisation Category 1 (UN-GHS) Cases of sensitisation also observed in humans. Not classified for respiratory sensitization
diethyleneglycol monomethacrylate	May cause sensitization by skin contact. Not classified for respiratory sensitization
Ethylene glycol dimethacrylate	Local Lymph Node Assay (LLNA), Mouse, May cause sensitization by skin contact. Not classified for respiratory sensitization

Carcinogenicity

Product:	Based on available data, the classification criteria are not met. No indications of critical properties in analogy to similar products or on the basis of structure-activity relationships. no specific test data available
Components:	
2-hydroxyethyl methacrylate	Not classified no specific test data available no evidence for hazardous properties (structure-activity-relationships) (analogy)
diethyleneglycol monomethacrylate	Not classified
Ethylene glycol dimethacrylate	Not classified

Germ Cell Mutagenicity

Based on available data, the classification criteria are not met.

In vitro

Product:	positive and negative test results
Components:	
2-hydroxyethyl methacrylate	gene mutation, OECD 476: , negative, CHO-cells (analogy) Cytogenetic Test (chromosome aberration), OECD 473: , positive, Chinese hamster lung cells gene mutation, OECD 471: , negative Based on available data, the classification criteria are not met.
diethyleneglycol monomethacrylate	Not classified No known chronic or acute health risks.
Ethylene glycol dimethacrylate	Cytogenetic Test (chromosome aberration), OECD 473: , positive, human lymphocytes

In vivo

Product:	no evidence of mutagenic effects
Components:	
2-hydroxyethyl methacrylate	Micronucleus test, OECD 474, Rat, male, negative
diethyleneglycol monomethacrylate	No information about adverse effects due to exposure.
Ethylene glycol dimethacrylate	Micronucleus test, OECD 474, Oral, Mouse, negative DNA damage and/or repair, Oral, Rat, male, negative

Reproductive toxicity

Product:	Animal experiments give no indications of reproduction-toxic effects
Components:	
2-hydroxyethyl methacrylate	Not classified No indications of toxic effects were observed in reproduction studies in animals.
diethyleneglycol monomethacrylate	Not classified
Ethylene glycol dimethacrylate	Not classified

Specific Target Organ Toxicity - Single Exposure

Product:	Based on available data, the classification criteria are not met.
Components:	
2-hydroxyethyl methacrylate	Not classified, no evidence for hazardous properties

diethyleneglycol monomethacrylate	Not classified
Ethylene glycol dimethacrylate	Not classified

Specific Target Organ Toxicity - Repeated Exposure

Product:	Based on available data, the classification criteria are not met.
Components:	
2-hydroxyethyl methacrylate	Not classified, no evidence for hazardous properties
diethyleneglycol monomethacrylate	Not classified
Ethylene glycol dimethacrylate	Not classified

Aspiration Hazard

Product:	No aspiration toxicity classification
Components:	
2-hydroxyethyl methacrylate	Not classified
diethyleneglycol monomethacrylate	Not classified
Ethylene glycol dimethacrylate	Not classified

Information on health hazards

Other hazards

Product:	Avoid contact with the skin and eyes and inhalation of the product vapours.

12. Ecological information

Ecotoxicity:

Toxicity to Aquatic Plants

Product:	EC 50, Selenastrum capricornutum (green algae), 72 h, 345 mg/l, OECD 201 NOEC, Selenastrum capricornutum (green algae), 72 h, 160 mg/l, OECD 201
Components:	
2-hydroxyethyl methacrylate	EC 50, Selenastrum capricornutum (green algae), 72 h, 345 mg/l, OECD 201 NOEC, Selenastrum capricornutum (green algae), 72 h, 160 mg/l, OECD 201
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	EC 50, Raphidocelis subcapitata (freshwater green alga), 72 h, 17,3 mg/l, OECD Test Guideline 201 EC 10, Raphidocelis subcapitata (freshwater green alga), 72 h, 6,93 mg/l, OECD Test Guideline 201, growth rate

Toxicity to microorganisms

Product:	EC 50, Pseudomonas fluorescens, 16 h, > 3.000 mg/l, Own test result.
Components:	
2-hydroxyethyl methacrylate	EC 50, Pseudomonas fluorescens, 16 h, > 3.000 mg/l, DEV L8, Own study

diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	EC 50, Pseudomonas putida, 3 h, 570 mg/l, OECD 209

Acute hazards to the aquatic environment:

Fish

Product:	LC 50, Oryzias latipes (Orange-red killifish), 96 h, > 100 mg/lOECD Test Guideline 203
Components:	
2-hydroxyethyl methacrylate	LC 50, Oryzias latipes (Orange-red killifish), 96 h, > 100 mg/lOECD Test Guideline 203
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	LC 50, Brachydanio rerio (zebrafish), 96 h, 15,95 mg/lOECD Test Guideline 203

Aquatic Invertebrates

Product:	EC 50, Daphnia magna (Water flea), 48 h, 380 mg/lOECD 202 part 1
Components:	
2-hydroxyethyl methacrylate	EC 50, Daphnia magna (Water flea), 48 h, 380 mg/lOECD 202 part 1
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	EC 50, Daphnia magna (Water flea), 48 h, 44,9 mg/lOECD 202

Chronic hazards to the aquatic environment:

Fish

Product:	Trial not required for scientific reasons.
Components:	
2-hydroxyethyl methacrylate	Trial not required for scientific reasons.
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	Trial not required for scientific reasons.

Aquatic Invertebrates

Product:	NOEC, Daphnia magna (Water flea), 21 Days, 24,1 mg/l, OECD 202 part 2
Components:	
2-hydroxyethyl methacrylate	NOEC, Daphnia magna (Water flea), 21 d, 24,1 mg/l, OECD 202 part 2
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	NOEC, Daphnia magna (Water flea), 21 d, 5,05 mg/l, OECD 211

Persistence and Degradability

Biodegradation

Product:	>= 92 %, 14 d, OECD 301 C, Readily biodegradable
Components:	

2-hydroxyethyl methacrylate	>= 92 %, 14 d, OECD 301 C, Readily biodegradable
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	69 %, 28 d, OECD 301 F, The 10 day time window criterion is not fulfilled., Readily biodegradable

BOD/COD Ratio

Product:	No data available.
Components:	
2-hydroxyethyl methacrylate	No data available.
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	No data available.

Bioaccumulative potential

Bioconcentration Factor (BCF)

Product:	Accumulation in organisms is not expected due to the coefficient of distribution of n-octanol in water (log Pow).
Components:	
2-hydroxyethyl methacrylate	Accumulation in organisms is not expected due to the coefficient of distribution of n-octanol in water (log Pow).
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	Accumulation in organisms is not expected due to the coefficient of distribution of n-octanol in water (log Pow).

Partition Coefficient n-octanol / water (log Kow)

Product:	0,42, 25 °C, OECD 107
Components:	
2-hydroxyethyl methacrylate	0,42, 25 °C, OECD Test Guideline 107
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	2,4, OECD Test Guideline 117

Mobility in soil:

Product:	Very sparingly volatile from the aqueous phase. Binding to the solid soil phase, sediment or clarification sludge is not expected.
Components:	
2-hydroxyethyl methacrylate	Very sparingly volatile from the aqueous phase. Binding to the solid soil phase, sediment or clarification sludge is not expected.
diethyleneglycol monomethacrylate	No data available.
Ethylene glycol dimethacrylate	Very sparingly volatile from the aqueous phase. The substance is distributed mainly into the water phase and the soil.

Results of PBT and vPvB assessment:

Product:	This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

2-hydroxyethyl methacrylate	Non-classified vPvB substance, Non-classified PBT substance
diethyleneglycol monomethacrylate	Non-classified vPvB substance, Non-classified PBT substance
Ethylene glycol dimethacrylate	Non-classified vPvB substance, Non-classified PBT substance

Other adverse effects:

Other hazards

Product: Prevent substance from entering soil, natural bodies of water and sewer systems. Photochemical degradation (air) takes place.

13. Disposal considerations

General information: Dispose of waste and residues in accordance with local authority requirements.

Disposal methods: Waste is hazardous. It must be disposed of in accordance with the regulations after consultation of the competent local authorities and the disposal company in a suitable and licensed facility. Strictly controlled conditions during disposal or treatment to air, wastewater and waste. Do not add wastewater to a biological wastewater treatment plant Bring wastewater containing AOX for professional disposal. The waste key number must be determined as per the European Waste Types List (decision on EU Waste Types List 2000/532/EC) in cooperation with the disposal firm / producing firm / official authority.

Contaminated Packaging: Contaminated packages must be emptied as good as possible. They may then be recycled after proper cleaning. Packages that cannot be cleaned must be disposed of in the same way as the substance. Packaging that cannot be cleaned should be disposed of professionally. Uncontaminated packaging may be taken for recycling.

14. Transport information

International Regulations

UNRTDG
Not regulated as a dangerous good

IATA-DGR
Not regulated as a dangerous good

IMDG-Code
Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

15. Regulatory information

International regulations

Montreal protocol
Not applicable

Stockholm convention
Not applicable

Rotterdam convention
Not applicable

Kyoto protocol
Not applicable

16.Other information, including date of preparation or last revision

Version #: 6.1

Generation date: 19.03.2025

Date of first report version: 10.03.2019

Abbreviations and acronyms:

AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CERCLA - Comprehensive Environmental Response, Compensation, and Liability Act; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DOT - Department of Transportation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; EHS - Extremely Hazardous Substance; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; HMIS - Hazardous Materials Identification System; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MSHA - Mine Safety and Health Administration; n.o.s. - Not Otherwise Specified; NFPA - National Fire Protection Association; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; RCRA - Resource Conservation and Recovery Act; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RQ - Reportable Quantity; SADT - Self-Accelerating Decomposition Temperature; SARA - Superfund Amendments and Reauthorization Act; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

Source of information:

relevant manuals and publications
own examinations
own toxicological and ecotoxicological studies
toxicological and ecotoxicological studies of other manufacturers
SIAR
OECD-SIDS
RTK public files

Further Information: The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.

Disclaimer: This information and all further technical advice is based on our present knowledge and experience. However, it implies no liability or other legal responsibility on our part, including with regard to existing third party intellectual property rights, especially patent rights. In particular, no warranty, whether express or implied, or guarantee of product properties in the legal sense is intended or implied. We reserve the right to make any changes according to technological progress or further developments. The customer is not released from the obligation to conduct careful inspection and testing of incoming goods. Performance of the product described herein should be verified by testing, which should be carried out only by qualified experts in the sole responsibility of a customer. Reference to trade names used by other companies is neither a recommendation, nor does it imply that similar products could not be used.